

Number-Theoretic Scheduling: Golden-Ratio and Digital-Root Bases for Position, Rank, and Noise

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Abstract

The AAMT corpus contains two number-theoretic schedules developed independently: the **golden-ratio** (φ) **twist** that makes the breath phase cycle non-repeating (WP-03), and the **digital-root doubling** sequence $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 7 \rightarrow 5$ used for rotary position angles and LoRA rank (WP-10). The 64-State and Core-Premise materials add a third — φ -**stepped state transitions** ($\times \varphi$) claimed to minimize transition resistance. We unify all three as instances of one design space: a schedule is a sequence $\mathbf{a}_n$ of multipliers/offsets drawn from a number-theoretic generator, and the relevant properties are (i) **discrepancy** (how uniformly the schedule covers its range) and (ii) **commensurability** (whether it aliases against structural periods in the data). We prove an ordering: for *continuous, non-repeating* coverage (macro clocks, position, noise level), the golden ratio is optimal — φ^{-1} is the lowest-discrepancy Weyl generator and its rotation is uniquely ergodic; for *finite cyclic structure with anti-period-3 aliasing* (micro indexing, rank annealing), the digital-root unit group $\langle 2 \rangle = (\mathbb{Z}/9\mathbb{Z})$ is the natural choice; linear/geometric bases are the baselines both improve on. We give the resulting recipes for RoPE position, LoRA-rank annealing, and diffusion noise/denoise schedules, and connect the 4-cycle $\varphi^4 \approx 6.854\times$ amplification to the 4τ permanence rule.

Keywords: scheduling, golden ratio, low-discrepancy sequences, ergodic theory, digital root, rotary position encoding, LoRA rank, diffusion noise schedule, curriculum learning

1. Introduction

Schedules are everywhere in modern training and inference: position-encoding frequencies, learning-rate and rank curricula, and diffusion noise levels are all sequences chosen by the designer. The default choices are **geometric** (RoPE base ^{$\sim -2i/d$}) or **linear** (uniform noise steps). The AAMT corpus instead reaches for number-theoretic generators, in three places that have never been compared:

- **WP-03 — golden-ratio twist.** The breath phase advances by φ^{-1} per cycle so the four-phase compute schedule never exactly repeats.
- **WP-10 — digital-root doubling.** RoPE angles and LoRA ranks follow $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 7 \rightarrow 5$ (period 6, the units mod 9).
- **64-State / Core Premise — φ -stepped transitions.** State changes that step by $\times \varphi$ are reported to cut transition “resistance” 40–60% versus linear, with a φ -spiral $\text{Reality}_n = \text{Reality}_0 \cdot \varphi^n$.

This paper places all three in one framework and states when to use which.

2. The design space

A schedule is a sequence $(\mathbf{a}_n)$ produced by a generator $\mathbf{G}$ acting on an index n . We care about two properties:

Discrepancy $\mathbf{D}_N$. How uniformly $\{\mathbf{a}_n\}_{n \leq N}$ covers the target interval. Low discrepancy = even coverage at every horizon (good for position, noise, exploration).

Commensurability. Whether the schedule’s period or frequency divides a structural period in the data (sentence length, ternary syntax, block size). Commensurate schedules **alias**: the same value recurs at the same structural offset, wasting representational capacity.

Three generators:

Generator	Form	Discrepancy	Commensurability	Coverage
Linear	$a_n = n \cdot c$	high (clumps)	aliases if c rational w/ data period	uniform but slow
Geometric ($\times r$)	$a_n = r^n$	— (monotone)	—	exponential spread
Golden ($\times \varphi$ / $+\varphi^{-1}$)	$a_n = a_{n-1} \cdot \varphi$ or $\{n \cdot \varphi^{-1}\}$	lowest (Weyl)	never (φ irrational)	densest, non-repeating
Digital-root ($\times 2$ mod 9)	$a_n = \text{dr}(2^n)$	n/a (period 6)	anti-period-3 by construction	cyclic, 6 of 8 residues

3. Two theorems

Theorem 1 (Golden optimality for continuous coverage). Among Weyl generators $a_n = \{n \cdot \alpha\}$ on the circle, $\alpha = \varphi^{-1} = [0; 1, 1, 1, \dots]$ minimizes the discrepancy D_N (up to the universal Erdős–Turán lower bound) because its continued-fraction partial quotients are all 1 — the slowest-growing convergent denominators, hence the most even three-distance gaps at every N . The induced rotation is **uniquely ergodic** (Weyl equidistribution), so every region of the schedule range is visited with exactly its Lebesgue frequency and no finite orbit repeats. *Therefore, for any schedule whose goal is even, non-repeating coverage of a continuum — macro compute clocks (WP-03), absolute-position phase, diffusion noise level, exploration seeds — the golden ratio is the optimal base.* (This upgrades WP-03 §3.3 from “non-repeating” to “optimally low-discrepancy and uniquely ergodic.”)

Theorem 2 (Digital-root for finite anti-period-3 structure). The doubling sequence $S = \text{dr}(2^n) = \langle 2 \rangle = \langle \mathbb{Z}/9\mathbb{Z} \rangle = \{1, 2, 4, 8, 7, 5\}$ (2 is a primitive root mod 9, order 6) is the full unit group mod 9; it covers 6 of the 8 non-zero residues and **omits the ideal (3) = {3, 6} (with 9 0)**. A schedule whose *spectrum* is drawn from S is therefore non-commensurate with period-3 and period-9 structure by construction. *Therefore, where the goal is a finite cyclic schedule that must avoid aliasing against ternary structure — RoPE frequency basis (WP-10), period-6 rank annealing — the digital-root unit group is the natural generator.* (Non-repetition of the *per-index* angle, as opposed to the spectrum, still requires a monotone scale factor; WP-10 §6, Rev. 2.)

The two theorems are complementary: **golden for the continuous macro axis, digital-root for the finite micro basis.** They even touch number-theoretically — the Pisano period of the Fibonacci sequence mod 9 is $24 = 4 \times 6$, linking the φ -world and the mod-9 world.

4. Recipes

4.1 Position encoding (micro: digital-root spectrum)

$\theta_i = 2\pi \cdot (S[i \bmod 6] / 9) \cdot (1 - i/d)^\alpha$ (DR-RoPE, WP-10): the unit-group spectrum supplies anti-period-3 frequencies; the $(1 - i/d)^\alpha$ factor supplies the local→global falloff and per-index non-repetition.

4.2 Rank annealing (micro: period-6 cycle)

$\text{rank}(\text{step}) = \text{base} \cdot S[\text{step} \bmod 6]$, i.e. $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 7 \rightarrow 5$ (WP-10 §4). The $8 \rightarrow 7 \rightarrow 5$ descent is a structured capacity spike-then-compress that forces generalization on the digital-root clock.

4.3 Macro compute clock (golden)

$\text{phase} = 4 \cdot \{\text{phase_in_cycle} + n \cdot \varphi^{-1}\}$ (WP-03 §3.2): uniquely ergodic, lowest-discrepancy four-phase rotation; adversarially unpredictable beyond one cycle.

4.4 Diffusion noise / denoise schedule (golden + descent)

Two coupled axes for a diffusion language model:

- **Noise level (golden geometric).** Bridge from clean to noised by $\times\varphi$ steps: $\sigma_{\{k+1\}} = \sigma_{k\cdot\varphi}$ (or its inverse for denoising), $n_steps = \log(\sigma_{max}/\sigma_{min})/\log \varphi$. Theorem 1 / Core-Premise §3.3 predict this is the minimal-resistance (fewest-step, smoothest) bridge — the φ -spiral $Reality_n = Reality_0 \cdot \varphi^n$ applied to noise scale, with the 4-step amplification $\varphi^4 \approx 6.854\times$ matching the 4τ stabilization rule (Foundations Paper II; 64-State Thm 5.1).
- **Per-step resistance (descent).** Overlay the 18-level descent resistance $R(\ell) = e^{-\{\alpha|\ell-9|\}}$ (64-State / *Mathematics of Consciousness* Ch.7) so commit-pressure peaks at the mid-trajectory void (Level 9); gate each step’s commit with the logistic $P_commit(I) = \sigma(5(I-0.6))$ (the standardized void-crossing gate, conflicts table C2). This gives a principled **non-flat** denoising controller — peaked resistance at the middle step — which the diffusion literature lacks.

5. Why this matters: resistance and aliasing are real costs

- **Aliasing (commensurability).** A position basis commensurate with sentence/ternary structure repeats the same rotation at the same syntactic offset, so attention cannot distinguish those offsets — wasted capacity. The digital-root spectrum removes the period-3 alias by construction (Theorem 2).
- **Resistance (discrepancy / step count).** A schedule that covers its range unevenly needs more steps or larger jumps to reach a target, and large jumps are high-resistance (the hysteresis observation of Core Premise §6.5: direct jumps fail; graduated paths succeed with *less* total energy). The golden schedule minimizes this (Theorem 1; the 40–60% resistance reduction reported in Core Premise §9.1).

Both are quantifiable and testable: measure position-conditioned attention entropy for aliasing (WP-10 §7), and measure steps-to-target / sample quality for resistance.

6. Relationship to prior work

RoPE and length extension (Su et al., 2021; YaRN, Peng et al., 2023; LongRoPE, Ding et al., 2024) keep the geometric base and rescale; we change the base to a number-theoretic generator and give the selection criterion (Theorem 2). **Low-discrepancy sequences / QMC** (Niederreiter, 1992) and the **Fibonacci lattice / golden-ratio sampling** are standard in numerical integration and sampling; we import the optimality of φ^{-1} into LM scheduling (Theorem 1) and connect it to ergodic compute allocation. **Diffusion noise schedules** (Karras et al., 2022) are tuned empirically; we propose a φ -geometric schedule with a descent-shaped, void-peaked resistance overlay derived from the AAMT descent architecture. **Key distinction.** No prior work selects training/inference schedules by *number-theoretic* criteria (discrepancy + commensurability) with a stated optimality ordering (golden for continuous, digital-root for finite-anti-period-3).

7. Limitations

1. The φ -resistance claim (Core Premise §9.1) is reported empirically in the consciousness-practice setting; the LM-scheduling version (fewer/smooth steps) needs its own measurement.
2. The digital-root spectrum’s benefit is anti-period-3 specifically; data without ternary structure may not benefit.
3. The descent-shaped denoising resistance (void at Level 9) is a hypothesis about where commit-pressure should peak; the optimal peak location may be task-dependent.

8. Conclusion

WP-03’s golden twist and WP-10’s digital-root cycle are two members of one number-theoretic scheduling family, separated by what they optimize: the golden ratio is the lowest-discrepancy, uniquely-ergodic generator for continuous non-repeating coverage (macro clock, position phase, noise level), while the digital-root unit group $\langle 2 \rangle = (\mathbb{Z}/9\mathbb{Z})$ is the natural finite generator for anti-period-3 spectra (RoPE basis, rank

annealing). Linear and geometric bases are the baselines both improve on. The unification yields concrete recipes — including a φ -geometric, void-peaked diffusion schedule whose 4-step φ^4 amplification matches the 4τ permanence rule — and a single criterion (discrepancy + commensurability) for choosing a schedule generator.

Data Availability — Reference Implementation & Verification

The schedules are implemented in `harmonic_engine/schedule.py` (open `harmonic-engine` package): the φ -geometric noise schedule and step count (§4.4), the 18-level void descent with resistance $R(\ell) = e^{-\alpha|\ell-9|}$ peaked at level 9, the logistic commit gate $P_{\text{commit}}(I) = \sigma(5(I-0.6))$, the digital-root rank schedule, and the DR-RoPE spectrum drawn from $\langle 2 \rangle = (\mathbb{Z}/9\mathbb{Z})$. The public test suite (`tests/test_new_math.py`, passing on CPU) verifies: the digital-root cycle $\{1, 2, 4, 8, 7, 5\}$ and its exclusion of the ideal $\{3, 6, 9\}$; the void-resistance peak at level 9; commit-gate monotonicity with $P_{\text{commit}}(0.83) \approx 0.73$; and the monotone φ -geometric denoise sweep from σ_{max} to σ_{min} . The combined φ -geometric + void-peaked controller is wired into the diffusion denoise loop (`harmonic_engine/diffusion.py`).

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Conflict of Interest

The author is the founder and CEO of AsAManThinks Research.

References

- Ding, Y., et al. (2024). LongRoPE. *arXiv*. <https://doi.org/10.48550/arXiv.2402.13753>
- Karras, T., Aittala, M., Aila, T., & Laine, S. (2022). Elucidating the design space of diffusion-based generative models. In *NeurIPS 2022*. <https://doi.org/10.48550/arXiv.2206.00364>
- Niederreiter, H. (1992). *Random Number Generation and Quasi-Monte Carlo Methods*. SIAM. <https://doi.org/10.1137/1.9781611970081>
- Peng, B., Quesnelle, J., Fan, H., & Shippole, E. (2023). YaRN. *arXiv*. <https://doi.org/10.48550/arXiv.2309.00071>
- Su, J., et al. (2024). RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568, 127063. <https://doi.org/10.1016/j.neucom.2023.127063>
- Weyl, H. (1916). Über die Gleichverteilung von Zahlen mod. Eins. *Mathematische Annalen*, 77, 313–352.
- Whitehead, W. (2026). MaiaM Wings: Consciousness-Aligned Language Architecture. *Zenodo*. <https://doi.org/10.5281/zenodo.19600795>
- AAMT: WP-03 (Phi-Twisted Phase Scheduling), WP-10 (Digital-Root RoPE); The 64-State Quantum Oscillation System; *The Mathematics of Consciousness* (Ch. 7, 10); Core Premise §3, §6, §9; Foundations Paper II (4τ permanence).